RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE
EXPERIMENT 5: INFORMATION EXTRACTION USING spaCy

QUESTION:
An e-commerce company receives thousands of customer reviews containing valuable information about products, brands, organizations, locations, and quantities. Develop an information extraction system using spaCy to preprocess customer reviews, identify tokens, lemmas, and part-of-speech tags, extract named entities, and display the extracted information in a structured format.

AIM:
To implement information extraction using spaCy on the Amazon Fine Food Reviews dataset by performing text preprocessing, tokenization, lemmatization, part-of-speech tagging, named entity recognition, and displaying the extracted information in a structured DataFrame.

ALGORITHM:
1. Import the required Python libraries, including Pandas and spaCy, and load the English language model.
2. Load the Amazon Fine Food Reviews dataset, remove missing and duplicate reviews, and retain the first 10,000 reviews.
3. Apply the spaCy NLP pipeline to five sample reviews and display the original text, tokens, lemmas, and part-of-speech tags.
4. Identify nouns, verbs, adjectives, and adverbs, and count the frequency of major part-of-speech categories.
5. Apply Named Entity Recognition (NER) to extract entities such as persons, organizations, locations, dates, quantities, and products where recognized by the model.
6. Create a function that accepts a customer review and returns its tokens, lemmas, POS tags, named entities, and entity labels.
7. Test the function using at least three new customer reviews.
8. Display the extracted information in a structured format using a Pandas DataFrame.

PROGRAM & OUTPUT:

```
# Name: Shyam M (241801267)
# Section 1: Import Libraries and Load spaCy Model

import pandas as pd
import spacy
from collections import Counter

!python -m spacy download en_core_web_sm

nlp = spacy.load("en_core_web_sm")
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 81.4 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
# Name: Shyam M (241801267)
# Section 2: Load and Clean Dataset

df = pd.read_csv("/content/Reviews.csv",
                 engine="python", on_bad_lines="skip")

df = df.dropna(subset=["Text"])
df = df.drop_duplicates(subset=["Text"])
df = df.head(10000).copy()

print("Dataset shape:", df.shape)
print(df[["Text"]].head())
```

```
Dataset shape: (6239, 10)
                                                Text
0  I have bought several of the Vitality canned d...
1  Product arrived labeled as Jumbo Salted Peanut...
2  This is a confection that has been around a fe...
3  If you are looking for the secret ingredient i...
4  Great taffy at a great price.  There was a wid...
```

```
# Name: Shyam M (241801267)
# Section 3: Tokenization, Lemmatization and POS Tagging

for review in df["Text"].head(5):
    doc = nlp(review)

    print("\nOriginal Review:", review[:200])
    print("Tokens:", [token.text for token in doc])
    print("Lemmas:", [token.lemma_ for token in doc])
    print("POS Tags:", [(token.text, token.pos_) for token in doc])
```

```
Original Review: I have bought several of the Vitality canned dog food products and have found them all to be of good q
Tokens: ['I', 'have', 'bought', 'several', 'of', 'the', 'Vitality', 'canned', 'dog', 'food', 'products', 'and', 'have',
Lemmas: ['I', 'have', 'buy', 'several', 'of', 'the', 'Vitality', 'can', 'dog', 'food', 'product', 'and', 'have', 'find'
POS Tags: [('I', 'PRON'), ('have', 'AUX'), ('bought', 'VERB'), ('several', 'ADJ'), ('of', 'ADP'), ('the', 'DET'), ('Vit

Original Review: Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not
Tokens: ['Product', 'arrived', 'labeled', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanuts', 'were', 'actuall
Lemmas: ['product', 'arrived', 'label', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanut', 'be', 'actually',
POS Tags: [('Product', 'NOUN'), ('arrived', 'AUX'), ('labeled', 'VERB'), ('as', 'ADP'), ('Jumbo', 'PROPN'), ('Salted',

Original Review: This is a confection that has been around a few centuries.  It is a light, pillowy citrus gelatin with
Tokens: ['This', 'is', 'a', 'confection', 'that', 'has', 'been', 'around', 'a', 'few', 'centuries', '.', ' ', 'It', 'is
Lemmas: ['this', 'be', 'a', 'confection', 'that', 'have', 'be', 'around', 'a', 'few', 'century', '.', ' ', 'it', 'be',
POS Tags: [('This', 'PRON'), ('is', 'AUX'), ('a', 'DET'), ('confection', 'NOUN'), ('that', 'PRON'), ('has', 'AUX'), ('b

Original Review: If you are looking for the secret ingredient in Robitussin I believe I have found it.  I got this in a
Tokens: ['If', 'you', 'are', 'looking', 'for', 'the', 'secret', 'ingredient', 'in', 'Robitussin', 'I', 'believe', 'I',
Lemmas: ['if', 'you', 'be', 'look', 'for', 'the', 'secret', 'ingredient', 'in', 'Robitussin', 'I', 'believe', 'I', 'hav
POS Tags: [('If', 'SCONJ'), ('you', 'PRON'), ('are', 'AUX'), ('looking', 'VERB'), ('for', 'ADP'), ('the', 'DET'), ('sec

Original Review: Great taffy at a great price.  There was a wide assortment of yummy taffy.  Delivery was very quick.
Tokens: ['Great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'There', 'was', 'a', 'wide', 'assortment', 'of', 'yum
Lemmas: ['great', 'taffy', 'at', 'a', 'great', 'price', '.', ' ', 'there', 'be', 'a', 'wide', 'assortment', 'of', 'yumm
POS Tags: [('Great', 'ADJ'), ('taffy', 'NOUN'), ('at', 'ADP'), ('a', 'DET'), ('great', 'ADJ'), ('price', 'NOUN'), ('.',
```

```
# Name: Shyam M (241801267)
# Section 4: POS Category Frequency

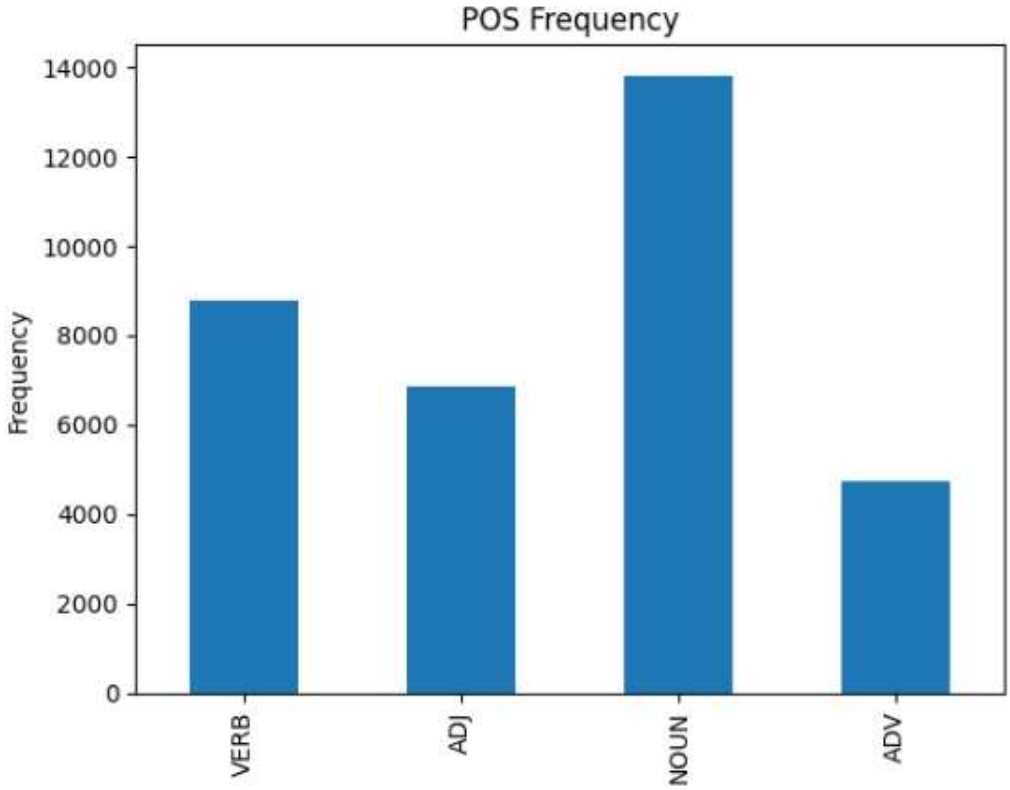
pos_counts = Counter()

for review in df["Text"].head(1000):
    doc = nlp(review)
    pos_counts.update(
        token.pos_ for token in doc
        if token.pos_ in ["NOUN", "VERB", "ADJ", "ADV"]
    )

print("\nPOS Category Frequencies:")
print(pos_counts)

pd.Series(pos_counts).plot(kind="bar", title="POS Frequency")
import matplotlib.pyplot as plt
plt.ylabel("Frequency")
plt.show()
```

```
POS Category Frequencies:
Counter({'NOUN': 13808, 'VERB': 8763, 'ADJ': 6832, 'ADV': 4709})
```




```python
# Name: Shyam M (241801267)
# Section 5: Named Entity Recognition

entity_records = []

for review in df["Text"].head(100):
    doc = nlp(review)

    for ent in doc.ents:
        entity_records.append({
            "Review": review[:100],
            "Entity": ent.text,
            "Entity Label": ent.label_
        })

entity_df = pd.DataFrame(entity_records)
print("\nExtracted Named Entities:")
print(entity_df.head(20))
```

```
Extracted Named Entities:
                                                 Review  \
0   I have bought several of the Vitality canned d...
1   I have bought several of the Vitality canned d...
2   Product arrived labeled as Jumbo Salted Peanut...
3   Product arrived labeled as Jumbo Salted Peanut...
4   This is a confection that has been around a fe...
5   This is a confection that has been around a fe...
6   This is a confection that has been around a fe...
7   This is a confection that has been around a fe...
8   This is a confection that has been around a fe...
9   This is a confection that has been around a fe...
10  This is a confection that has been around a fe...
11  This is a confection that has been around a fe...
12  If you are looking for the secret ingredient i...
13  If you are looking for the secret ingredient i...
14  Great taffy at a great price.  There was a wid...
15  I got a wild hair for taffy and ordered this f...
16  I got a wild hair for taffy and ordered this f...
17  This saltwater taffy had great flavors and was...
18  Right now I'm mostly just sprouting this so my...
19  I don't know if it's the cactus or the tequila...

                              Entity Entity Label
0                           Vitality          ORG
1                         My Labrador       PERSON
2                        Jumbo Salted          ORG
3                               Jumbo  WORK_OF_ART
4            around a few centuries          DATE
5            pillowy citrus gelatin          ORG
6                           Filberts          GPE
7                       C.S. Lewis'          ORG
8              The Lion, The Witch  WORK_OF_ART
9                           Edmund      PRODUCT
10                         Sisters       PERSON
11                           Witch          ORG
12                       Robitussin          GPE
13          the Root Beer Extract I          LAW
14                        Delivery       PERSON
15                       five pound     QUANTITY
16                   only two weeks          DATE
17                      Fralinger's          ORG
18                       Wheatgrass       PERSON
19  Tequila Picante Gourmet de Inclan          ORG
```

```python
# Name: Shyam M (241801267)
# Section 6: Create Information Extraction Function

def extract_information(review):
    doc = nlp(review)

    return {
        "Original Review": review,
        "Tokens": [t.text for t in doc],
        "Lemmas": [t.lemma_ for t in doc],
        "POS Tags": [(t.text, t.pos_) for t in doc],
        "Named Entities": [
            (ent.text, ent.label_) for ent in doc.ents
        ]
    }
```

```python
# Name: Shyam M (241801267)
# Section 7: Test with Three Customer Reviews

reviews = [
    "This chocolate product from Nestle tastes excellent.",
    "I received the package in Chennai yesterday.",
    "The product costs 20 dollars and arrived quickly."
]

results = [extract_information(review) for review in reviews]

result_df = pd.DataFrame(results)
print("\nStructured Extraction Results:")
print(result_df.to_string(index=False))
```

```
Structured Extraction Results:
                                  Original Review                                                                  Tokens
This chocolate product from Nestle tastes excellent. [This, chocolate, product, from, Nestle, tastes, excellent, .] [th
       I received the package in Chennai yesterday.         [I, received, the, package, in, Chennai, yesterday, .]
   The product costs 20 dollars and arrived quickly.   [The, product, costs, 20, dollars, and, arrived, quickly, .]
```

RESULT:

Thus, information extraction was successfully performed on the Amazon Fine Food Reviews dataset using spaCy. The customer reviews were processed to identify tokens, lemmas, and part-of-speech tags. Named Entity Recognition was used to extract entities such as persons, organizations, locations, dates, and quantities. The extracted information was organized into a structured DataFrame for further analysis of customer feedback.